A residual replacement for the linear incidence guide

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Abstract

Stochastic epidemic models of cumulative incidence are used when new infections are recorded in discrete windows and the latent path is not observed. A linear guide proposes the latent counts by freezing the hazard until the next observation and shifting it with a Gaussian approximation of the count. The importance weight keeps the discretised transition in the numerator. On repeated synthetic series the median effective sample size of this guide falls from 149.66 at the published spacing to 2.06 when the spacing is five times coarser. A myopic proposal does not fall in the same way. We replace the Poisson means by the increment of a restarted linear-noise path, together with a conditional residual mean for infections. The denominator remains a product of Poisson probabilities for the same increments. At the coarse spacing the residual median effective sample size is 169.45. The residual median still declines as the spacing grows, and at the coarse spacing the gain is concentrated in the second window. In five of 20 replicates every draw in the first window has weight zero. The study uses a synthetic susceptible-infected-removed mechanism.

Keywords. conditioned hazard, effective sample size, Markov jump process, particle filter, SIR model

1 Introduction

Cumulative incidence is a common observation in epidemic modelling when infection times are hidden and only counts of new cases are recorded in successive windows. Whitaker et al. (2025) take a time-discretised susceptible-exposed-infected-removed model, and its susceptible-infected-removed reduction, as the latent transmission process, and they assimilate noisy incidence with a particle filter. The same construction matters more widely for

partially observed jump processes, where a conditioned proposal is used because forward simulation ignores the next observation (Golightly and Wilkinson 2015; Golightly and Sherlock 2019).

The inferential obstacle is an intractable observed-data likelihood. The data are incomplete and discrete, the reporting model is noisy, and the latent incidence between two observation times is a path of counts. A bootstrap filter would draw that path from the unconditioned transition and weight by the observation alone. Whitaker et al. (2025, p. 8) give two reasons not to stop there. Resampling static parameters collapses them, and forward simulation of incidence ignores the next count.

Whitaker et al. (2025) meet the second difficulty with a conditioned hazard. Particles are propagated by replacing the Poisson rate with a hazard derived from a Gaussian approximation of the cumulative incidence between the current time and the next observation, given that observation. The particle is then reweighted by the discretised transition and the observation mass function. The limitations section of that paper states that a further assumption in the derivation is linearity of the latent process over an interval no longer than the observation interval. When the epidemic is sparsely observed, so that cumulative incidence does not evolve linearly, "the conditioned hazard is likely to result in trajectories that are inconsistent with the true conditioned process" (Whitaker et al. 2025, p. 25).

That paragraph gives no numerical spacing at which the failure begins, and it does not say that the importance weight then targets a different distribution. It names recent work on sampling conditioned jump processes, in particular Golightly and Sherlock (2019) and Corstanje et al. (2023), as relevant and open to adaptation. Neither algorithm is printed there as a replacement inside the filter.

Our contribution is to replace the Poisson means inside that proposal and to measure the effect on the weights. We treat the constant-hazard step in the 2025 derivation as the linearity assumption named on that page, so that the inconsistency described there is a property of the proposal. We report the effective sample size of the incidence weights on the synthetic susceptible-infected-removed mechanism of Whitaker et al. (2025), at the published spacing and at two coarser spacings fixed in advance. Equation (19) of Golightly and Sherlock (2019) and Theorem 4.2 of Corstanje et al. (2023) describe proposals for objects other than the incidence increments in the filter. Section 5 states the replacement we use: the increment of a restarted incidence linear-noise path, plus a conditional residual mean for infections. The density in the denominator remains a product of Poisson probabilities for those incidence increments.

The numerical illustration uses the synthetic mechanism in Section 4.1 of Whitaker et al. (2025). That paper also discusses weekly Ebola counts and weekly COVID-19 counts. The comparison in this paper is confined to the synthetic mechanism.

The remainder of this paper is organised as follows. Section 2 compares related guides. Section 3 restates the 2025 filter and the linear incidence guide. Section 4 states the diagnostic. Section 5 states the residual proposal. Section 6 reports both comparisons, and Section 7 discusses their scope.

2 Related work

We compare guides by what they propagate, what they condition on, and whether an importance weight corrects the proposal. Table 1 summarises the constructions used below.

Equation (9) of Golightly and Wilkinson (2015) is the constant-hazard step that Whitaker et al. (2025) adapt. The observation in the 2015 paper is a Gaussian function of the state, and the observation in the incidence filter is a binomial or negative binomial count. Equation (5) of Golightly and Sherlock (2019) restates that hazard. Their Section 3.2 says that normality of the remaining reaction counts, and a hazard that is constant on the whole remaining interval, are often unreasonable. Their Section 6 says that treating the time until the next observation as one discretisation makes the bridges less accurate as the inter-observation time increases. Equation (19) of the same paper is a separate proposal: a jump-process bridge under a Gaussian observation of the state, with weight (3). Its density is the jump-process likelihood ratio, and the incidence filter requires a product of Poisson probabilities on a grid.

Theorem 4.2 of Corstanje et al. (2023) conditions an inhomogeneous Poisson process on an exact integer end-point. The likelihood ratio (16) contains $h(0, x_0)$, the transition probability of the original process. Their Section 4.1 does not treat a binomial observation of an increment on a grid. The proposal in Section 5 uses the incidence linear-noise equations of Golightly et al. (2023) and the residual covariance block of Whitaker et al. (2017, Section 3.1.2), and it evaluates a Poisson probability for the same increments as the numerator.

Lowe et al. (2023) accelerate particle Markov chain Monte Carlo for a chemical kinetic model. Their conditioned hazard multiplies a jump hazard by a ratio of observation densities, and they replace that density by a linear-noise Gaussian. The construction follows Golightly and Sherlock (2019),

and it differs from the incidence proposal in Whitaker et al. (2025). They remark that nonlinear behaviour between observations precludes a modified diffusion bridge that pushes particles towards the observations in a linear fashion. For that bridge they name Durham and Gallant (2002), and for a residual bridge they name Whitaker et al. (2017). The residual covariance in Section 5 is the block in Section 3.1.2 of the latter paper.

Jovanovski et al. (2024) build data-conditional proposals for approximate Bayesian computation on a diffusion. Their discussion of erratic paths is a conjecture about where their speed-up is largest, and it concerns their own proposals. Whitaker et al. (2025, pp. 2–3) also cite Golightly and Kypraios (2018). Equation (6) of that paper is the constant-hazard correction adapted by the 2025 guide. Section 5 uses a different mean.

3 The 2025 filter and the linear guide

Consider the filter of Whitaker et al. (2025). Let X_t denote prevalence and let N_t denote cumulative incidence. The unconditioned hazard is h, and $H = \mathrm{diag}\{h\}$. A scalar observation y_{t_i} is a noisy count of one incidence component over $(t_{i-1}, t_i]$, selected by a vector P. In the susceptible-infected-removed applications, $P' = (1, 0)$. The contact rate enters on the log scale as $\tilde{\beta}_t$, in the precision form printed as equation (2) of Whitaker et al. (2025). The static parameter and the particle index are those of their Algorithm 2. The comparison below holds the rate parameters fixed, so the gamma updates of their equations (9)–(11) stay outside the calculation.

Their Algorithm 2 is printed for the negative binomial observation. The synthetic study in their Section 4.1 uses the binomial observation, and the observation factor in the weight is the binomial probability mass function.

Algorithm 1. Schematic of Algorithm 2 of Whitaker et al. (2025). The draw from q in the propagation step is the linear incidence guide. Section 5 replaces the Poisson means of that draw.

1. Propagate each particle. Jitter the observation parameter with the kernel printed for the negative binomial model. Draw the contact-rate path from its transition, which the paper calls a blind propagation. Draw the incidence path from the proposal q, which depends on the next observation.

2. Resample with the weight in Equation (1).

3. Update the sufficient statistic for the rate parameters.

4. Draw those rate parameters from their conditional posteriors.

The weight in step 2, for particle k on the window that ends at the next observation, is

$$w^{(k)} \propto \frac{\pi\big(\Delta n^{(k)} \mid \tilde{\beta}^{(k)}, \theta^{(k)}\big)\, \pi\big(y \mid \Delta n^{(k)}, \phi^{(k)}\big)}{q\big(\Delta n^{(k)} \mid \tilde{\beta}^{(k)}, \phi^{(k)}, \theta^{(k)}, y\big)}. \tag{1}$$

The numerator is the time-discretised transition and the observation mass function. It does not contain the conditioned hazard h^*. The denominator is q. We use the ratio as an importance weight for one propagate step.

The proposal q is built in their Section 3.4. At a time τ inside the observation window, equation (12) of Whitaker et al. (2025) approximates the remaining incidence by a Gaussian whose mean and variance are the current hazard times the remaining time. The observation is given a Gaussian approximation, and the mean inside the observation variance is replaced by the mean of that Gaussian. The display after their equation (18) is the conditioned hazard h^*. End-point paths are their forward recursion with h replaced by h^*. The density q is the product of Poisson probabilities with rates $h^*\Delta\tau$.

The printed correction multiplies by $\rho P' H$, a row when the observation is scalar. The cross-covariance block that enters the conditional mean of the remaining incidence is a column. For a diagonal H the row and the column have the same entries. On the susceptible-infected-removed model the shared entries correct the infection component only,

$$h^*_{\mathrm{inf}} = h_{\mathrm{inf}} + \rho h_{\mathrm{inf}} \frac{y - \hat{\mu}}{\rho^2 h_{\mathrm{inf}}(t_i - \tau) + \sigma^2}, \qquad h^*_{\mathrm{rem}} = h_{\mathrm{rem}}, \tag{2}$$

where $\hat{\mu} = \rho\big(n^{\mathrm{past}}_{\mathrm{inf}} + h_{\mathrm{inf}}(t_i - \tau)\big)$ and, for the binomial variance evaluated at $\hat{\mu}$, $\sigma^2 = (1-\rho)\hat{\mu}$. Equation (2) is the scalar form used in the simulation. Its entries match the printed display, and they match equation (9) of Golightly and Wilkinson (2015) when the stoichiometry is the identity, so that the correction is the column HP.

The study in Section 6 holds the contact-rate path and the prevalence at the left endpoint of the window at the simulated values, and it holds the removal rate and the reporting probability at the values used to simulate the data. The jitter, the blind contact-rate draw, the sufficient-statistic update and the conditional parameter draw are omitted. A particle in Algorithm 1 is propagated without the latent path. The comparison concerns h^* given that path.

4 Diagnostic

Equation (12) of Whitaker et al. (2025) freezes the hazard at the left endpoint of the remaining interval. The mean of the remaining incidence is that frozen hazard times the remaining time. Golightly and Wilkinson (2015, Section 3.1) use the same constant-hazard Gaussian step for remaining reaction counts. Golightly and Sherlock (2019, Section 3.2) say that this step is often unreasonable, and their Section 6 says that accuracy falls as the inter-observation time grows. Whitaker et al. (2025, p. 25) call linearity of the latent process, over an interval no longer than the observation interval, a further assumption in the derivation of the conditioned hazard. Their Section 3.4 uses neither the word linear nor a second condition beyond equation (12) and the display after (18). We take the constant-hazard step to be the assumption the limitations paragraph is pointing at.

The same paragraph says the conditioned hazard is likely to propose trajectories inconsistent with the true conditioned process, when cumulative incidence does not evolve linearly. The law named on the page is the true conditioned process, and the sentence is about the trajectories the hazard proposes. The discretised transition in the numerator of Equation (1) stays in the importance target, which for one propagate step is that transition times the observation mass function. This is a reading of the printed weight.

A poor h^* changes q and therefore the weights. The limitations paragraph gives no numerical spacing at which the 2025 guide fails for this epidemic model. Golightly and Sherlock (2019, Section 5.1) report that a linear form fails to describe exponential decay, and that effective sample size falls as the interval length grows, for a death process. Those figures concern a death process, and the comparison below is for the epidemic model.

The observable is the effective sample size of the unnormalised incidence weights in one window,

$$\mathrm{ESS}(\tilde{w}_{1:N}) = \frac{\left(\sum_{i=1}^{N} \tilde{w}_i\right)^2}{\sum_{i=1}^{N} \tilde{w}_i^2}, \tag{3}$$

as printed by Golightly and Sherlock (2019, Section 5.2). Zero weights are included. If every weight in a window is zero, Equation (3) is undefined, and the value entered in the table is 0. A draw has weight zero when the correction in Equation (2) is undefined, when a component of the hazard passed to the Poisson draw is negative, or when the updated counts would be negative. Whitaker et al. (2025) print no truncation of h^*. Golightly and Sherlock (2019, Section 3.2) truncate their equation (5) to keep it positive.

A negative hazard in the 2025 display is left untruncated and receives weight zero.

The comparison, specified before any weight was computed, is between the median of this effective sample size across replicate data sets at the coarsest spacing and the same median at the published unit spacing. A myopic proposal, with h^* replaced by h, is computed on the same paths as a comparator. The diagnostic is that comparison across the three spacings. It supplies no threshold that would divide adequate spacing from inadequate spacing, and it leaves the filtering distribution unexamined. Section 5 states the proposal used in the second comparison.

5 Residual incidence guide

The residual guide changes the Poisson means and keeps the Poisson draw, so that q is the product of Poisson probabilities on the same incidence increments that appear in the numerator of Equation (1). The numerator itself is unchanged.

The limitations paragraph of Whitaker et al. (2025) points to constructions that sample different objects. Equation (19) of Golightly and Sherlock (2019) samples a Markov jump process, and the weight (3) is the jump-process likelihood ratio under a Gaussian observation of the state. Theorem 4.2 of Corstanje et al. (2023) conditions an inhomogeneous Poisson process on an exact integer end-point, and the ratio (16) contains the transition probability $h(0, x_0)$. Corollary 4.3.6 and Theorem 4.4.2 of Corstanje (2024) are path-space ratios for a guided chemical reaction process conditioned on an exact end constraint. Equations (15) and (25) of Whitaker et al. (2017) are Gaussian densities of a diffusion skeleton. The denominator used here is the Poisson product described above, and the factor $h(0, x_0)$ remains inside ratio (16) wherever that ratio is discussed.

At a substep, the linear-noise approximation of cumulative incidence is restarted at the current prevalence. The incidence coordinate η starts at zero, the fundamental matrix G starts at the identity, and the residual variance V starts at zero. The contact rate is held at its current value. We integrate equations (4), (8) and (9) of Golightly et al. (2023) over the next substep and over the time remaining until the observation. The Jacobian is the matrix printed after their equation (6). The restart is the one they describe after their equation (19). Fearnhead et al. (2014, Section 4.2) state that restart: the ordinary differential equation is re-initialised at the beginning of each interval. The integration uses a classical Runge–Kutta method

with step length 0.1, chosen before the comparison in Section 6.

The mean of the binomial observation then uses the linear-noise mean of the remaining infections in place of the frozen product $h_{\mathrm{inf}}(t_i - \tau)$. The variance remains the binomial variance in Equation (2), $\sigma^2 = (1-\rho)\hat{\mu}$. For an observation of infections this is the variance in equation (22) of Golightly et al. (2023).

Let R denote the residual incidence about η, with variance V after the restart. The covariance between the residual at the next substep and the residual at the end of the remaining interval is $V_{\Delta\tau}(G_{t_i-\tau}G_{\Delta\tau}^{-1})'$. Whitaker et al. (2017, Section 3.1.2) write the same block as $P_t\psi_t P_T'$ after their display (24). The observation loads on the infection component by ρ. The conditional mean of the infection residual at the next substep is that infection covariance, times $\rho(y-\hat{\mu})$, divided by $\rho^2 V_{11}(t_i-\tau)+\sigma^2$. The removal mean is the ordinary-differential increment of the removal coordinate, in line with Equation (2), which corrects the infection component only.

The Poisson mean for the next infection increment is the ordinary-differential increment of η plus that conditional residual mean. The Poisson mean for the removal increment is the ordinary-differential increment of the removal coordinate alone. A draw receives weight zero if the integration stops because a prevalence on the ordinary differential equation would be negative, if the absolute value of the determinant of $G_{\Delta\tau}$ is smaller than 10^{-12}, if either Poisson mean is negative or is not finite, or if the updated counts would be negative. The truncation in Section 3.2 of Golightly and Sherlock (2019) applies to their equation (5), and a negative Poisson mean here is left unchanged.

Equation (6) of Golightly and Kypraios (2018) is the constant-hazard correction that Whitaker et al. (2025) adapt. The means in this section are a separate construction. Fintzi et al. (2022) reparameterise a linear-noise approximation for incidence counts. The equations integrated here are those printed by Golightly et al. (2023), who cite that reparameterisation.

6 Simulation

The study uses the synthetic susceptible-infected-removed mechanism of Section 4.1 of Whitaker et al. (2025). The initial state is $(s, i) = (762, 5)$, the removal rate is 0.5, the initial log contact rate is -6, the contact-rate precision is 100 and the reporting probability is 0.9, with a binomial observation. The latent path is simulated with time step 0.001 on the interval from 0 to 10, and the filter uses substeps of length 0.1. Observations are

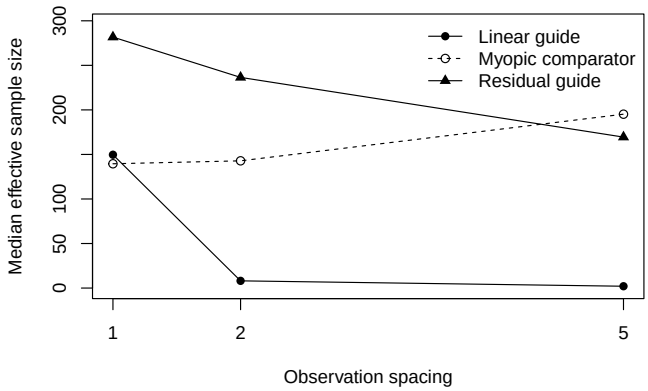


Figure 1: Effective sample size against observation spacing. Filled circles: linear guide, the medians in Table 2. Open circles: myopic comparator. Triangles: residual guide, the medians in Table 3. The residual comparison at spacing 5 is with the linear median.

taken at spacings 1, 2 and 5. Spacing 1 is the published grid. Spacings 2 and 5 are coarser, and both were fixed before any weight was computed. All algorithms were coded in R. The reported summaries are effective sample sizes of the proposal weights. The design does not target a posterior distribution of the static parameters, so there is no gold-standard posterior.

We drew 20 replicates, with 500 proposals in each window. Replicate r was initialised with seed $20251015 + r$, for $r = 1, \ldots, 20$. Binomial observations at spacings 1, 2 and 5 were drawn in that order, before the proposals. Every simulated path was retained.

For each replicate, each spacing and each guide, the within-replicate summary is the median of the window values of Equation (3). The reported summary is the median of those 20 numbers. For the linear guide the comparison is between spacing 5 and spacing 1. The myopic guide uses the same simulated paths and the same observations, with independent Poisson draws. Table 2 reports the linear guide. Figure 1 plots the linear medians, the myopic medians and the residual medians from Table 3.

From Table 2, the linear median effective sample size is 149.66 at spacing 1 and 2.06 at spacing 5. The coarse spacing gives the smaller value, which is the comparison specified above. At spacing 2 the linear median is 8.09. At every linear row, the median that includes zero weights equals the median

computed on the finite weights alone, so the fall is present in the finite weights. The median proportion of invalid draws is 0.033 at spacing 1, 0.009 at spacing 2 and 0 at spacing 5.

The myopic median is 139.53 at spacing 1, 142.82 at spacing 2 and 195.18 at spacing 5. The myopic median increases with the spacing.

Two qualifications belong with the comparison. The guide received the simulated prevalence and the simulated contact-rate path, so the result describes h^* given those paths. At spacing 1 the linear median is 149.66 out of 500 draws, and the guide is already uneven on the published grid. The comparison is across spacings. It measures no pathwise distance to the true conditioned process.

The residual guide of Section 5 used the same mechanism, the same seeds and the same observation draws. The linear proposals from the first comparison were kept. The ordinary differential equations were integrated with a Runge–Kutta step of length 0.1. The comparison, specified before these weights were computed, asks whether the residual median effective sample size at spacing 5 exceeds the linear median 2.06. With two windows, the within-replicate median is the average of the two window values, the same functional as the linear summary at spacing 5.

From Table 3, the residual median is 281.71 at spacing 1, 236.47 at spacing 2 and 169.45 at spacing 5. The spacing-5 value exceeds 2.06. The residual median declines as the spacing grows, and it remains above the linear value. At spacing 5 the median on the finite weights is 173.33. The median proportion of invalid draws is 0.029 at spacing 1, 0.010 at spacing 2 and 0.0005 at spacing 5.

That spacing-5 proportion is a median across replicates, and the two windows behave differently. Five of the 20 first windows have all 500 residual draws invalid. The median effective sample size of the 20 first windows is 5.87, and the median of the 20 second windows is 331.99. The comparison with 2.06 uses the average of the two windows. On the same spacing the linear guide has median effective sample size 1.36 on the first window and 1.62 on the second, and every linear window has a positive effective sample size. The residual median at spacing 5 therefore combines a weak first window, covering the rise of the outbreak, with a much stronger second window.

7 Discussion

We have given a reading of the linear incidence guide in Whitaker et al. (2025) and a comparison of its weights under coarser observation. The constant-hazard step is the linearity assumption named in their limitations section. On the printed importance weight, a poor conditioned hazard changes the proposal and the weights. On the synthetic mechanism of their Section 4.1, the median effective sample size of the linear guide is 2.06 at spacing 5 and 149.66 at spacing 1 (Table 2). The myopic comparator increases over the same spacings. The guide was given the simulated state and the simulated contact-rate path, and the published spacing already yields a median well below the number of draws.

The comparison is useful when an observation window is lengthened and the question is whether this proposal is the source of uneven weights. A posterior for the static parameters is outside its scope. The three spacings were chosen before the comparison because they divide the published horizon, and they are a design choice rather than estimated change-points.

The residual guide in Section 5 replaces the Poisson means inside that proposal. The new means are the increment of a restarted incidence linear-noise path, plus a conditional residual mean on the infection count. The denominator of Equation (1) remains the product of Poisson probabilities for those increments. The median effective sample size at spacing 5 is 169.45, set against the linear median 2.06. The same summary declines from 281.71 at spacing 1. The gain at spacing 5 is uneven across the two windows. The first window, which covers the rise of the outbreak, has median effective sample size 5.87, and in five replicates every draw in that window has weight zero. The second window has median 331.99. A guide that improves the second half of a sparsely observed outbreak, and still fails on the first half, is only a partial replacement.

Equation (19) of Golightly and Sherlock (2019) samples a Markov jump process under a Gaussian observation of the state, with weight (3). Theorem 4.2 of Corstanje et al. (2023) conditions an inhomogeneous Poisson process on an exact integer end-point, and its ratio (16) contains $h(0, x_0)$. The density q in this paper is the Poisson product of Section 5.

The comparison uses the synthetic mechanism of Section 4.1 of Whitaker et al. (2025). The weekly Ebola series and the weekly COVID-19 series discussed in that paper would need the counts in reproducible form before they could be analysed in the same way.

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Disclosure statement

The authors report there are no competing interests to declare.

Data availability

All data used in this paper were simulated from the model stated in Section 6, with the settings given there.

Supplementary material

The online supplement contains a base R implementation of the linear guide and the residual guide, together with the code that reproduces the simulation in Section 6.

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Table 1: Guides compared in this paper. The first row is the proposal under study.

Guide	What is propagated	What the proposal conditions on	Weight
Whitaker et al. (2025), display after (18)	Poisson incidence increments on a grid, at rate $h^*\Delta\tau$	The next incidence observation, through a Gaussian approximation	Yes. The ratio uses the discretised transition and the observation mass function over q
Golightly and Wilkinson (2015), equation (9)	A Markov jump process, by exponential waits at the conditioned hazard	A Gaussian observation of the state. The hazard is frozen on the remaining interval	Yes. The complete-data likelihood ratio (10)
Golightly and Sherlock (2019), equation (5)	The same construction as Golightly and Wilkinson (2015), equation (9)	The same Gaussian observation. Section 3.2 says the constant-hazard step is often unreasonable	Yes, through the weight (3), when this hazard is the proposal
Golightly and Sherlock (2019), equation (19)	A Markov jump process, using the linear-noise conditioned hazard without restarting the ordinary differential equations	The same Gaussian observation, through the linear-noise transition density in their equation (6)	Yes. The weight is (3)
Corstanje et al. (2023), Theorem 4.2	An inhomogeneous Poisson process with a homogeneous-Poisson guide	An exact integer end-point	The ratio (16) is printed and contains the transition probability $h(0, x_0)$

Table 2: Linear incidence guide. Median effective sample size across 20 replicates, reported to two decimal places. Finite-weight medians use the draws with a finite log weight. Myopic medians are given in Section 6.

Spacing	ESS	ESS, finite weights	Invalid proportion
1	149.66	149.66	0.033
2	8.09	8.09	0.009
5	2.06	2.06	0

Table 3: Residual incidence guide. Median effective sample size across 20 replicates, reported to two decimal places. Finite-weight medians use the draws with a finite log weight. The comparison at spacing 5 is with the linear median 2.06 in Table 2.

Spacing	ESS	ESS, finite weights	Invalid proportion
1	281.71	281.71	0.029
2	236.47	236.47	0.010
5	169.45	173.33	0.0005